

6.3 注释

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2026 年 3 月 21 日

注释 1. 命题 3.4 (ii) $\Rightarrow$ (iii) 省略的步骤。

$\alpha = k\epsilon_i + \epsilon_j$ 代入上式,

$$\begin{aligned}(\mathcal{U}(k\epsilon_i + \epsilon_j), \mathcal{U}(k\epsilon_i + \epsilon_j)) &= (k\mathcal{U}\epsilon_i + \mathcal{U}\epsilon_j, k\mathcal{U}\epsilon_i + \mathcal{U}\epsilon_j) \\&= k(\mathcal{U}\epsilon_i, k\mathcal{U}\epsilon_i + \mathcal{U}\epsilon_j) + (\mathcal{U}\epsilon_j, k\mathcal{U}\epsilon_i + \mathcal{U}\epsilon_j) \\&= k(\overline{k\mathcal{U}\epsilon_i + \mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_i}) + (\overline{k\mathcal{U}\epsilon_i + \mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_j}) \\&= k(\overline{k\mathcal{U}\epsilon_i}, \overline{\mathcal{U}\epsilon_i}) + (\overline{\mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_i}) + (\overline{k\mathcal{U}\epsilon_i}, \overline{\mathcal{U}\epsilon_j}) + (\overline{\mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_j}) \\&= k\overline{k} + (\overline{\mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_i}) + \overline{k}(\overline{\mathcal{U}\epsilon_i}, \overline{\mathcal{U}\epsilon_j}) + 1 \\&= k[\overline{k} + (\overline{\mathcal{U}\epsilon_i}, \overline{\mathcal{U}\epsilon_j})] + \overline{k}(\overline{\mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_i}) + 1 \\&= k\overline{k} + k(\overline{\mathcal{U}\epsilon_i}, \overline{\mathcal{U}\epsilon_j}) + \overline{k}(\overline{\mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_i}) + 1\end{aligned}$$

又

$$\begin{aligned}(k\epsilon_i + \epsilon_j, k\epsilon_i + \epsilon_j) &= k(\epsilon_i, k\epsilon_i + \epsilon_j) + (\epsilon_j, k\epsilon_i + \epsilon_j) \\&= k(\overline{k\epsilon_i + \epsilon_j}, \overline{\epsilon_i}) + (\overline{k\epsilon_i + \epsilon_j}, \overline{\epsilon_j}) \\&= k(\overline{k\epsilon_i}, \overline{\epsilon_i}) + (\overline{\epsilon_j}, \overline{\epsilon_i}) + (\overline{k\epsilon_i}, \overline{\epsilon_j}) + (\overline{\epsilon_j}, \overline{\epsilon_j}) \\&= k\overline{k} + 1\end{aligned}$$

于是

$$\begin{aligned}k\overline{k} + k(\overline{\mathcal{U}\epsilon_i}, \overline{\mathcal{U}\epsilon_j}) + \overline{k}(\overline{\mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_i}) + 1 &= k\overline{k} + 1 \\k(\overline{\mathcal{U}\epsilon_i}, \overline{\mathcal{U}\epsilon_j}) + \overline{k}(\overline{\mathcal{U}\epsilon_j}, \overline{\mathcal{U}\epsilon_i}) &= 0\end{aligned}$$

注释 2. (ii) $\Rightarrow$ (iii) 的证明方法, 对正交变换是否成立, 即命题 2.1 中 (iv) 推 (iii)

设 $\mathcal{A}k\epsilon_i + \epsilon_j \in V (i \neq j)$,

$$\begin{aligned} (\mathcal{A}k\epsilon_i + \epsilon_j, \mathcal{A}k\epsilon_i + \epsilon_j) &= (k\mathcal{A}\epsilon_i + \mathcal{A}\epsilon_j, k\mathcal{A}\epsilon_i + \mathcal{A}\epsilon_j) \\ &= (k\mathcal{A}\epsilon_i, k\mathcal{A}\epsilon_i) + (k\mathcal{A}\epsilon_i, \mathcal{A}\epsilon_j) + (\mathcal{A}\epsilon_j, k\mathcal{A}\epsilon_i) + (\mathcal{A}\epsilon_j, \mathcal{A}\epsilon_j) \\ &= k \cdot k(\mathcal{A}\epsilon_i, \mathcal{A}\epsilon_i) + k(\mathcal{A}\epsilon_i, \mathcal{A}\epsilon_j) + k(\mathcal{A}\epsilon_j, \mathcal{A}\epsilon_i) + (\mathcal{A}\epsilon_j, \mathcal{A}\epsilon_j) \\ &= k \cdot k + 2k(\mathcal{A}\epsilon_i, \mathcal{A}\epsilon_j) + 1 \end{aligned}$$

又

$$\begin{aligned} (k\epsilon_i + \epsilon_j, k\epsilon_i + \epsilon_j) &= (k\epsilon_i, k\epsilon_i) + (k\epsilon_i, \epsilon_j) + (\epsilon_j, k\epsilon_i) + (\epsilon_j, \epsilon_j) \\ &= k \cdot k + 1 \end{aligned}$$

于是

$$\begin{aligned} k \cdot k + 2k(\mathcal{A}\epsilon_i, \mathcal{A}\epsilon_j) + 1 &= k \cdot k + 1 \\ (\mathcal{A}\epsilon_i, \mathcal{A}\epsilon_j) &= 0 \end{aligned}$$

又因为

$$\begin{aligned} (\mathcal{A}\epsilon_i, \mathcal{A}\epsilon_i) &= (\epsilon_i, \epsilon_i) \\ &= 1 \end{aligned}$$

所以 $\mathcal{A}\epsilon_1, \epsilon_2, \dots, \epsilon_n$ 是标准正交基。

注释 3. $(\mathcal{A}\alpha, \beta) = (\alpha, \mathcal{A}^*\beta)$ 显然等价于

$$(\mathcal{A}\epsilon_i, \epsilon_j) = (\epsilon_i, \mathcal{A}^*\epsilon_j) \quad (i, j = 1, 2, \dots, n)$$

的原因。

任意 $\alpha, \beta \in V$, 可表示为

$$\begin{aligned} \alpha &= x_1\epsilon_1 + x_2\epsilon_2 + \dots + x_n\epsilon_n \\ \beta &= y_1\epsilon_1 + y_2\epsilon_2 + \dots + y_n\epsilon_n \end{aligned}$$

于是

$$\begin{aligned}
(\mathcal{A}\alpha, \beta) &= (\mathcal{A}x_1\epsilon_1 + x_2\epsilon_2 + \cdots + x_n\epsilon_n, y_1\epsilon_1 + y_2\epsilon_2 + \cdots + y_n\epsilon_n) \\
&= (x_1\mathcal{A}\epsilon_1 + x_2\mathcal{A}\epsilon_2 + \cdots + x_n\mathcal{A}\epsilon_n, y_1\epsilon_1 + y_2\epsilon_2 + \cdots + y_n\epsilon_n) \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i (\mathcal{A}\epsilon_i, y_j\epsilon_j) \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i \overline{(y_j\epsilon_j, \mathcal{A}\epsilon_i)} \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i y_j \overline{(\epsilon_j, \mathcal{A}\epsilon_i)} \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i \overline{y_j} (\epsilon_j, \mathcal{A}\epsilon_i) \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i \overline{y_j} (\mathcal{A}\epsilon_i, \epsilon_j)
\end{aligned}$$

又

$$\begin{aligned}
(\alpha, \mathcal{A}\beta) &= (x_1\epsilon_1 + x_2\epsilon_2 + \cdots + x_n\epsilon_n, \mathcal{A}^*y_1\epsilon_1 + y_2\epsilon_2 + \cdots + y_n\epsilon_n) \\
&= (x_1\epsilon_1 + x_2\epsilon_2 + \cdots + x_n\epsilon_n, y_1\mathcal{A}^*\epsilon_1 + y_2\mathcal{A}^*\epsilon_2 + \cdots + y_n\mathcal{A}^*\epsilon_n) \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i (\epsilon_i, \mathcal{A}^*y_j\epsilon_j) \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i \overline{(y_j\mathcal{A}^*\epsilon_j, \epsilon_i)} \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i y_j \overline{(\mathcal{A}^*\epsilon_j, \epsilon_i)} \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i \overline{y_j} (\mathcal{A}^*\epsilon_j, \epsilon_i) \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i \overline{y_j} (\epsilon_i, \mathcal{A}^*\epsilon_j)
\end{aligned}$$

- 左边推右边

取 $x_1 = 1, y_1 = 1$, 其他坐标都为 0, 可得 $(\mathcal{A}\epsilon_1, \epsilon_1) = (\epsilon_1, \mathcal{A}^*\epsilon_1)$ 。
类似地, 推导出剩余项。

- 右边推左边
是平凡的结果。

综上，命题得证。

注释 4. 定理 3.2 推论中 U 的理解。

提示： U 是两个标准正交基之间的过渡矩阵。